Part 1a: Provide short answer only.

1. A mapping $F : \mathbf{R}^2 \to \mathbf{R}^3$, is given by the rule $F(x, y) = (x, y, y)$. Is F one-to-one, onto, expand or contract?

2. Let $a_1 = (1,2,3)$ and $a_2 = (4,-4,1)$, write a vector with minimum length that is a member of $\{a_1, a_2\}^{\perp}$

3. Consider a matrix $A = \begin{bmatrix} 2 & 1 & a \\ 0 & -3 & 6 \\ 0 & 0 & b \end{bmatrix}$, give one example value of both a and b that makes nullity(A) = 1

4. Let A be a matrix that represents a linear map whose domain is $\mathbb{R}^5$ and range is $\mathbb{R}^3$. Now consider a pair (rank(A),nullity(A)). How many different pairs are possible for this A. Note: a pair (x, y) is considered different from (p, q) when $x \neq p$ or $y \neq q$.

5. Let $y = mx + b$ be the line of best fit in the x-y plane through the points (-1,0), (0,0), (1,3). Find m and b.

Part 1b: Provide short answer only.

1. What is the angle between **v** and **w** when **v** = (1, 1, 0, 1, 1) and **w** = (1, 0, 0, 0, 1) in $\mathbf{R}^5$?

2. Find a linear equation whose solution set is the line through (2, 3) and perpendicular to (1, -2).

3. What are the number of columns and rows of a smallest matrix A such that $col(A) = span(\{(-1,0,1,0)\})$

4. Write a set P such that $span(P) = S \cap S^{\perp}$ where $S = \{(1,0,0,0)\}^{\perp}$

5. Let A be an $m \times n$ matrix such that the nullity of AA^T is c. Find rank(A) + dim(row(A)) + dim($N(A)$) in terms of m, n and c.

Part 1c: Provide short answer only.

1. Suppose we have geometric vectors ***a*** and ***p*** in space with numeric coordinates **a** = (1, 1, 1) and **p** = (2, -1, -3), respectively. What equation describes the plane ***P*** orthogonal to the line generated by ***a*** and containing the tip of ***p***?

2. Find an equation for the plane containing (1, 1, 1), (1, 2, 3) and (3, 5, 1).

3. Find three vectors on the plane 2x − 3y + 4z = 3.

4. Let $a_1 = (-1,0,0)$ and $a_2 = (0,1,0)$ and $a_3 = (x, y, z)$. We know that $\{a_1, a_2, a_3\}$ is an orthonormal basis of $\mathbb{R}^3$ and $x + y + z \neq 1$. Find a_3.

5. Consider a set $S = \{(2,6), (3,-5), (-3,-9), (-6,10)\}$. How many subsets of S are linearly independent? Note: there are 16 possible subsets of S.

Part 1d: Provide short answer only.

For problem 1-3, let A be a 2×2 matrix such that $A\begin{pmatrix}8\\6\end{pmatrix} = \begin{pmatrix}18\\24\end{pmatrix}$. Let $U\Sigma V^T$ be an svd of A such that the first column of U is $\begin{pmatrix}0.6\\0.8\end{pmatrix}$ and one of the singular value of A is 2.

1. Find the first column of V.
2. Find the determinant of A.
3. How many different matrix A are possible?

For problem 4-5, let $A = \begin{bmatrix}1 & 1 & 5\\0 & 3 & 2\\0 & 0 & 2\end{bmatrix}$.

4. Find the largest eigenvalue of A^3.
5. Find the eigenvector of A^3 corresponding with the largest eigenvalue.

Part 2a: Provide detailed answers.

2-1 Find all solutions of the system using the Gauss-Jordan elimination algorithm.

a + b + 2c = 1

2a – b + d = -2

a -b -c -2d = 4

2a -b + 2c -d = 0

2-2 Consider a matrix A where some entries are missing $A = \begin{bmatrix} -4 & 2 & -2 & 6 \\ 3 & -1 & 1 & 0 \\ 8 & ??? & 3 & ??? \end{bmatrix}$. We know that the basis of $row(A)$ contains exactly 2 vectors. Calculate followings set and explain how you calculate the result, i.e., describe why your calculation gives the required set. No point will be awarded if there is no explanation.

a. basis of $row(A)$
b. Smallest set S such that $S^{\perp} = nul(A)$
c. Smallest set T such that $span(T) = col(A)$
d. Smallest set U such that $span(U) = nul(A^T)$

2-3 Find the characteristic polynomial, all the eigenvalues and its corresponding eigenvectors for the following matrix:

$$\begin{bmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ -3 & 4 & 6 \end{bmatrix}$$

Part 2b: Provide detailed answers.

2-4 Consider three planes in space that go through the origin and are respectively perpendicular to (1, 0, -1), (1, 1, 1), and (0, 1, -1). Determine whether these planes intersect in just the origin, an entire line, or a complete plane.

2-5 Let $h: \mathbb{R}^3 \to \mathbb{R}^2$ be a linear map and let A is a matrix that represents h. We know that $h\big((1,0,0)\big) = (-2,4)$ and know that $(-4,-2)$ is in the left nullspace of A. Let $B = \{b_1, b_2, b_3\}$ and $D = \{d_1, d_2\}$ be a basis of $\mathbb{R}^3$ and $\mathbb{R}^2$ and that B and D are orthogonal but not necessarily orthonormal. We also know that $b_3 = (1,0,0)$. Additionally, we know that $Rep_{B,D}(h) = \begin{bmatrix} -2 & 5 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Find A, B and D that together satisfy all the above information. Also explain how you come up with this solution. No point will be awarded if there is no explanation.

2-6 Let A be an $m \times n$ matrix and A^+ be its pseudoinverse. Describe the mapping given by AA^+ and A^+A. Explain what they do to the related subspaces and how they can be useful.